

RoboBlast Grenade Verifier

Status: PASS
Godot: 4.6-stable (official)
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Source: ...0703-151656\reference-main-complete\evidence\score.json

Executive score report for deterministic grenade weapon benchmark output.

91/100

PASS

91 / 100

Total benchmark score

Passing threshold met.

The PDF mirrors the verifier JSON result. Category bars below show which behaviors were observed at runtime and which remain incomplete.

Category Scores

Category	Score	Observed
Weapon Controls	15/15	
Hud Feedback	10/10	
Trajectory Preview	24/30	
Projectile Physics	15/15	
Explosion Gameplay	17/20	
Visual Audio Polish	5/5	
Stability Repeatability	5/5	

Key Findings

1. Trajectory Preview (24/30)	visible grenade aiming aid appears in grenade mode; aiming aid does not clearly communicate an arcing grenade throw; aiming aid updates after aim or camera direction changes; aiming aid direction matches the thrown grenade direction; aiming aid remains available in grenade mode and hides after leaving grenade mode
2. Weapon Controls (15/15)	a weapon-switch input route also carries a controller binding; weapon mode observably switches through swap_weapons action; attack after weapon switch created runtime nodes; rapid grenade attacks appear rate-limited; default aimed attack still creates runtime nodes
3. Hud Feedback (10/10)	player has visible UI controls; UI control state changed after weapon switch; aiming changes UI feedback
4. Projectile Physics (15/15)	grenade attack spawned a nearby 3D node; spawned node follows arcing motion; projectile path stayed clear of player body

5. Explosion Gameplay (17/20)	default throw calibration full at 11.20 units: default throw calibration measured 11.20 units from Grenade after 10.16 units of horizontal travel; nearby target damage averaged across explosion trials: Seed 17031 throw raw nearby hit score 5/10; Seed 27059 throw raw nearby hit score 10/10; Seed 37087 throw raw nearby hit score 10/10; nearby damageable-only destructible damage averaged across explosion trials: Seed 17031 throw nearby damageable-only destructible was not damaged; Seed 27059 throw nearby damageable-only destructible was not damaged; Seed 37087 throw nearby damageable-only destructible was damaged; blast locality averaged across explosion trials: Seed 17031 throw explosion damage stayed localized; Seed 27059 throw explosion damage stayed localized; Seed 37087 throw explosion damage stayed localized; all explosion safety trials protected out-of-range targets; player safety averaged across explosion trials: Seed 17031 throw player safe; Seed 27059 throw player safe; Seed 37087 throw player safe; detonation effects averaged across explosion trials: Seed 17031 throw detonation produced runtime nodes or effects; Seed 27059 throw detonation produced runtime nodes or effects; Seed 37087 throw detonation produced runtime nodes or effects; default throw calibration is full quality inside the 6-12 unit envelope
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Detailed Item Analysis

Weapon Controls Detail (15/15)

Result	Check	Evidence
Earned 2/2	Controller weapon-switch binding	a weapon-switch input route also carries a controller binding
Earned 2/2	Weapon switch input responds	weapon mode observably switches through swap_weapons action
Earned 5/5	Grenade attack after switching	attack after weapon switch created runtime nodes
Earned 3/3	Rapid attack rate limit	rapid grenade attacks appear rate-limited
Earned 3/3	Default weapon still works	default aimed attack still creates runtime nodes

Hud Feedback Detail (10/10)

Result	Check	Evidence
Earned 3/3	Visible UI controls	player has visible UI controls
Earned 4/4	Weapon-switch UI state	UI control state changed after weapon switch
Earned 3/3	Aiming UI feedback	aiming changes UI feedback

Trajectory Preview Detail (24/30)

Result	Check	Evidence
Earned 5/5	Visible grenade aiming aid	visible grenade aiming aid appears in grenade mode
Missed 0/6	Communicates arcing throw	aiming aid does not clearly communicate an arcing grenade throw
Earned 8/8	Updates with aim/camera direction	aiming aid updates after aim or camera direction changes
Earned 7/7	Preview matches projectile direction	aiming aid direction matches the thrown grenade direction
Earned 4/4	Visibility lifecycle/cooldown behavior	aiming aid remains available in grenade mode and hides after leaving grenade mode

Projectile Physics Detail (15/15)

Result	Check	Evidence
Earned 4/4	Projectile spawned	grenade attack spawned a nearby 3D node
Earned 8/8	Arcing motion	spawned node follows arcing motion
Earned 3/3	Player-safe path	projectile path stayed clear of player body

Explosion Gameplay Detail (17/20)

Result	Check	Evidence
Partial 4/5	Nearby target damage across angles	default throw calibration full at 11.20 units: default throw calibration measured 11.20 units from Grenade after 10.16 units of horizontal travel; nearby target damage averaged across explosion trials: Seed 17031 throw raw nearby hit score 5/10; Seed 27059 throw raw nearby hit score 10/10; Seed 37087 throw raw nearby hit score 10/10
Partial 1/3	Nearby destructible damage across angles	nearby damageable-only destructible damage averaged across explosion trials: Seed 17031 throw nearby damageable-only destructible was not damaged; Seed 27059 throw nearby damageable-only destructible was not damaged; Seed 37087 throw nearby damageable-only destructible was damaged
Earned 5/5	Blast locality across angles	blast locality averaged across explosion trials: Seed 17031 throw explosion damage stayed localized; Seed 27059 throw explosion damage stayed localized; Seed 37087 throw explosion damage stayed localized; all explosion safety trials protected out-of-range targets
Earned 2/2	Player safety across angles	player safety averaged across explosion trials: Seed 17031 throw player safe; Seed 27059 throw player safe; Seed 37087 throw player safe
Earned 3/3	Detonation effects across angles	detonation effects averaged across explosion trials: Seed 17031 throw detonation produced runtime nodes or effects; Seed 27059 throw detonation produced runtime nodes or effects; Seed 37087 throw detonation produced runtime nodes or effects
Earned 2/2	Throw distance calibration quality	default throw calibration is full quality inside the 6-12 unit envelope

Visual Audio Polish Detail (5/5)

Result	Check	Evidence
Earned 2/2	Visible effect nodes	visible grenade or explosion nodes appeared
Earned 2/2	Detonation audio	audio player was active during detonation window
Earned 1/1	Temporary node cleanup	some temporary nodes cleaned up

Stability Repeatability Detail (5/5)

Result	Check	Evidence
Earned 1/1	Repeated grenade attacks	two separated grenade attacks produced runtime behavior
Earned 1/1	Default attack after grenade use	default aimed attack still works after grenade use
Earned 1/1	Main scene loads	main scene loaded with a playable player
Earned 1/1	Main scene default attacks	main scene default shooting and melee both produced runtime behavior
Earned 1/1	Main scene actors and pickups	main scene kept targeteables, damageables, collect_coin, and damage-spawned pickup behavior